Intro to AI for Medicine

Lec 1: Intro

Welcome!

Today's Agenda

1. Logistics, mindset, goals
2. History of AI up to 1990

Brief lecture today – expect 45 min

Logistics - Format

- Lectures 2x weekly, Monday, Wednesday
- 7pm
- Always on zoom
- Office hours for 1 hour after class

Logistics - Grading

- Pass/fail class
- Taking attendance at each class via zoom records
- 70% is passing per SOM policy, so $\frac{x}{23} = 70\%, x = 16.1$
- You can miss 6 lectures and still pass.
  - Must attend 17
  - I encourage attendance of all of the first 6
- 50% attendance, 50% final project (will discuss details later)
  - Project will be designing plan for a real-life AI intervention of your own

Lecture Schedule

- Foundations Module: 8 lectures
  - Focuses on how AI works
  - Mathematical
  - Will always have a “clinical correlate”
- Applications Module: 15 lectures
  - Tech applications (LLMs, Computer Vision, Interpretability)
  - Practical applications (AI infrastructure, medicolegal concerns, business)
  - Ethical applications (fairness, bias, ethics, etc)
- Will sprinkle in general CS knowledge PRN

Homework

- Refer to the weekly schedule for readings

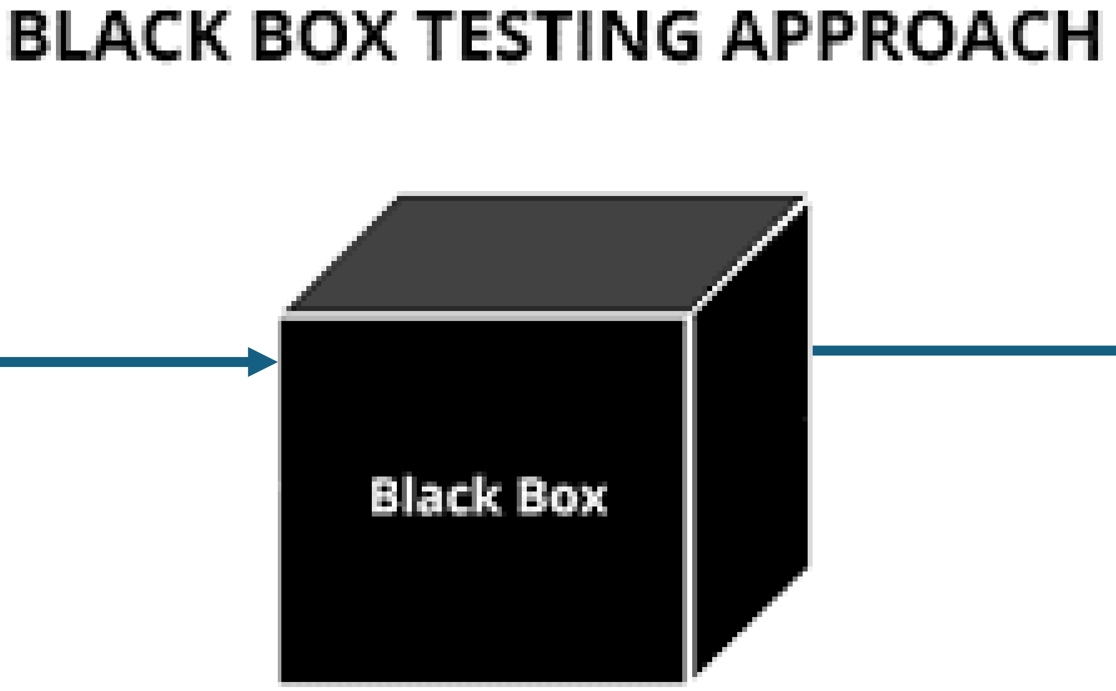
Lecture Number	Topics	Clinical Correlate	Assignments	Supplemental Material
1	Course intro, course structure, how to approach this class, what is AI, 4 pillars, formulation of AI problems,	Many	ELIZA – First Chatbot	
2	Necessary math for AI. What is a vector, matrix, tensor. matrix multiplication, inversion, transposition, points, surfaces and functions in n-dimensions, gradients, gradient descent. Includes brief history of modern AI.	PET/CT Tumor Detection	Matrix multiplication Matrix multiplication worksheet (until you get it)	Matrix multiplication demo
3	Loss functions. Linear regression as an example. Loss surfaces and gradient descent. Local and global minima. Mean squared error, cross entropy, domain specific loss.	Predicting hospital stay length	Google lec Gradient descent demo	MIT Lec 2 Linear regression (dense)
4	Data encodings. One hot, numerical, thermometer	Predictors of	Logistic	Logistic

Textbooks

- Math Textbook: MIT 6.036
- Ethics Textbook: FairML Book
  - **Barocas, Solon, Moritz Hardt, and Arvind Narayanan. *Fairness and Machine Learning: Limitations and Opportunities*. MIT Press, Dec. 2023.**
  - https://fairmlbook.org/

- No readings required, just a source for extra info if you want to dig deeper
- At times a source for supplemental material

Approach to this class



Patient has + swab for Gonorrhea

Treat with Ceftriaxone + doxycycline

But why?

Approach to this class

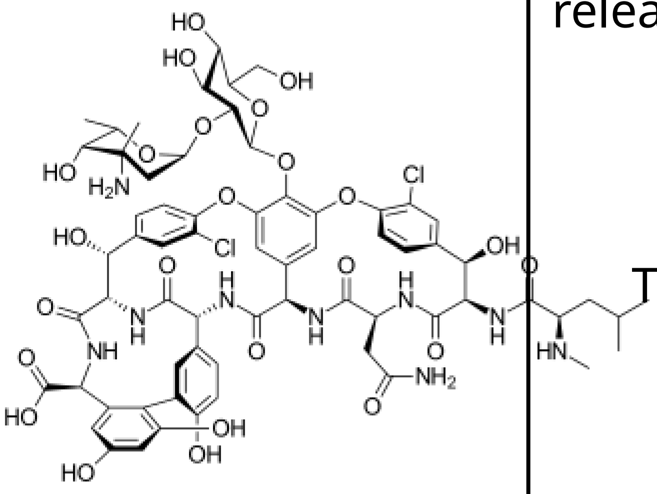
Look inside the box!

Chlamydia is intracellular – won't be detected unless released

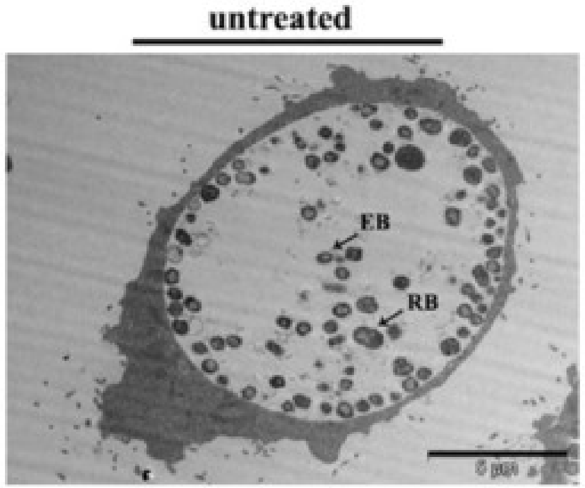
Treat like it's there

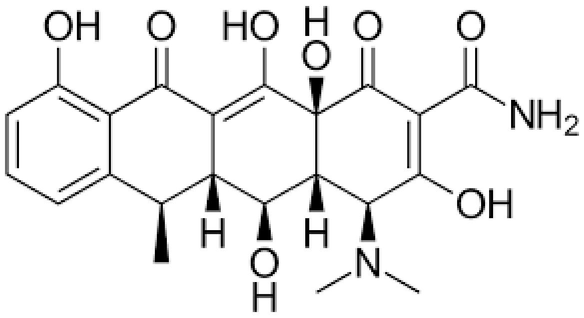
Treat with an antibiotic that's small and nonpolar, easily crosses cell membranes and can get intracellular

Treat chlamydia with doxycycline



Vancomycin

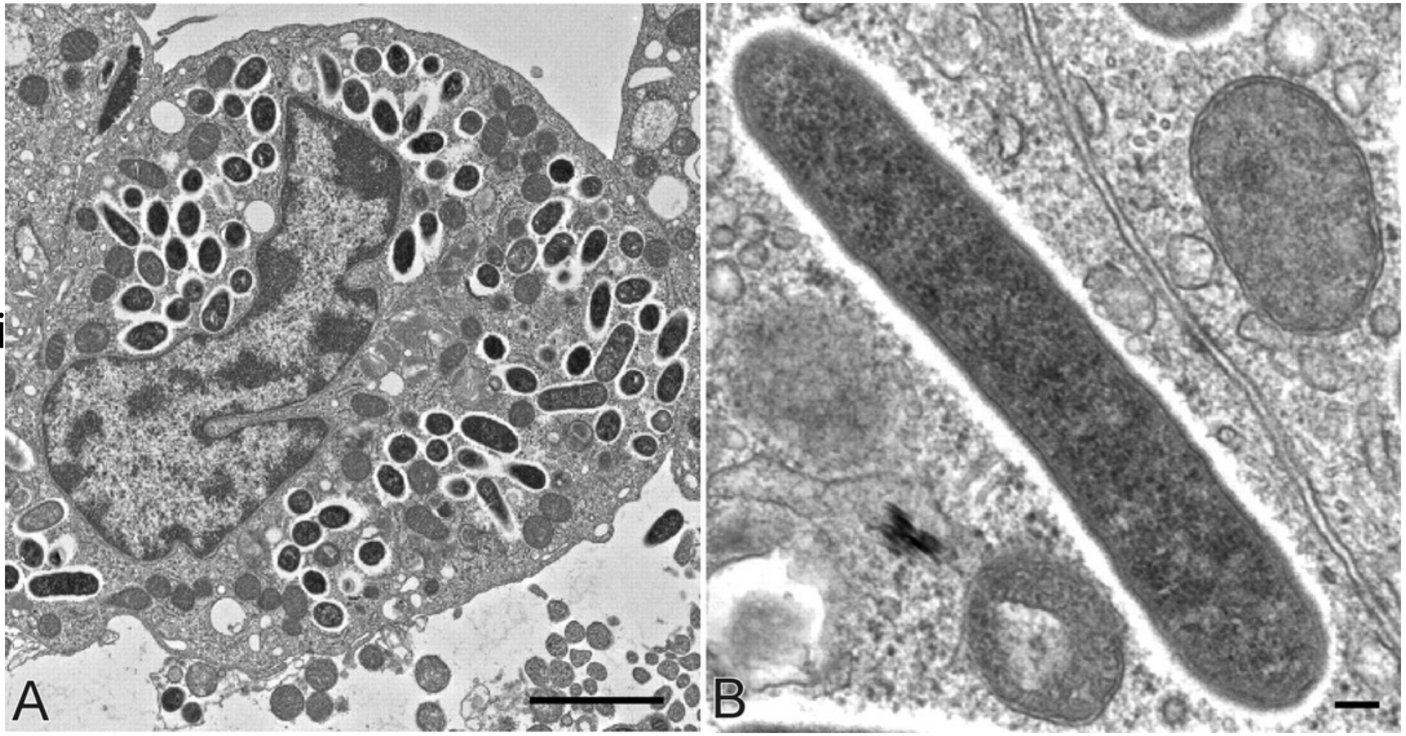




doxycycline

Approach to this class

Rickettsia Rickettsii



What would you treat with?

Treat RMSF with Doxycycline.

Approach to this class

- Give information to make decisions on your own
  - Who knows what technology will be out there in 2, 5, 10 years.
  - I want to give you tools to tackle new problems
- Can learn way more info in much less time
  - Sticks better too
- Teach you another way of thinking about problems, including in medicine

My Goals For The Class

01

Take it seriously and do a good job

- Make it meaningful to your future careers
- Adapt in real time to make this happen

02

Teach you a new way of approaching problems and thinking

03

Prepare you for any innovations that happen in your career

04

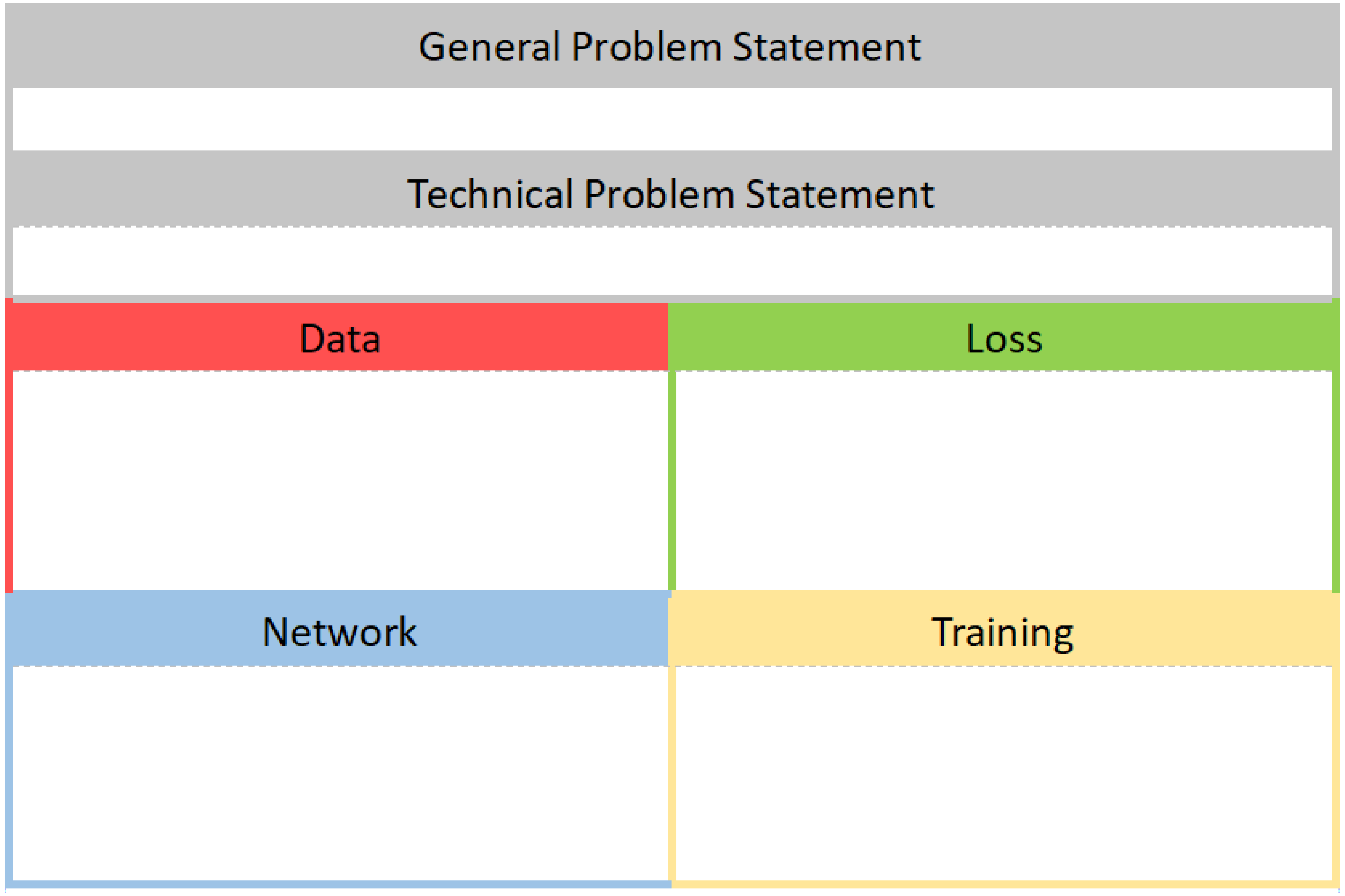
Break down difficult concepts into understandable words – this class should never feel overwhelming

05

Give frameworks and spot trends to give you intuition

You will be shocked how far we will get in 3 months!

4 Pillars Diagram



Your Goals Should Be

- Don't treat this class as a joke – try to learn something
  - Take breaks, absences as you need them
- Be open minded to learning from many disciplines
- Try to imagine how your practice and research can adapt
- Give feedback!

Tips for Success in This Class

- You get out what you put in
- This class is truly multidisciplinary, so learn to think like an engineer
  - Ask lots of questions!
- Give feedback in real time

What is AI?

What is AI? (Two More Definitions)

- Artificial intelligence (AI) is the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making.

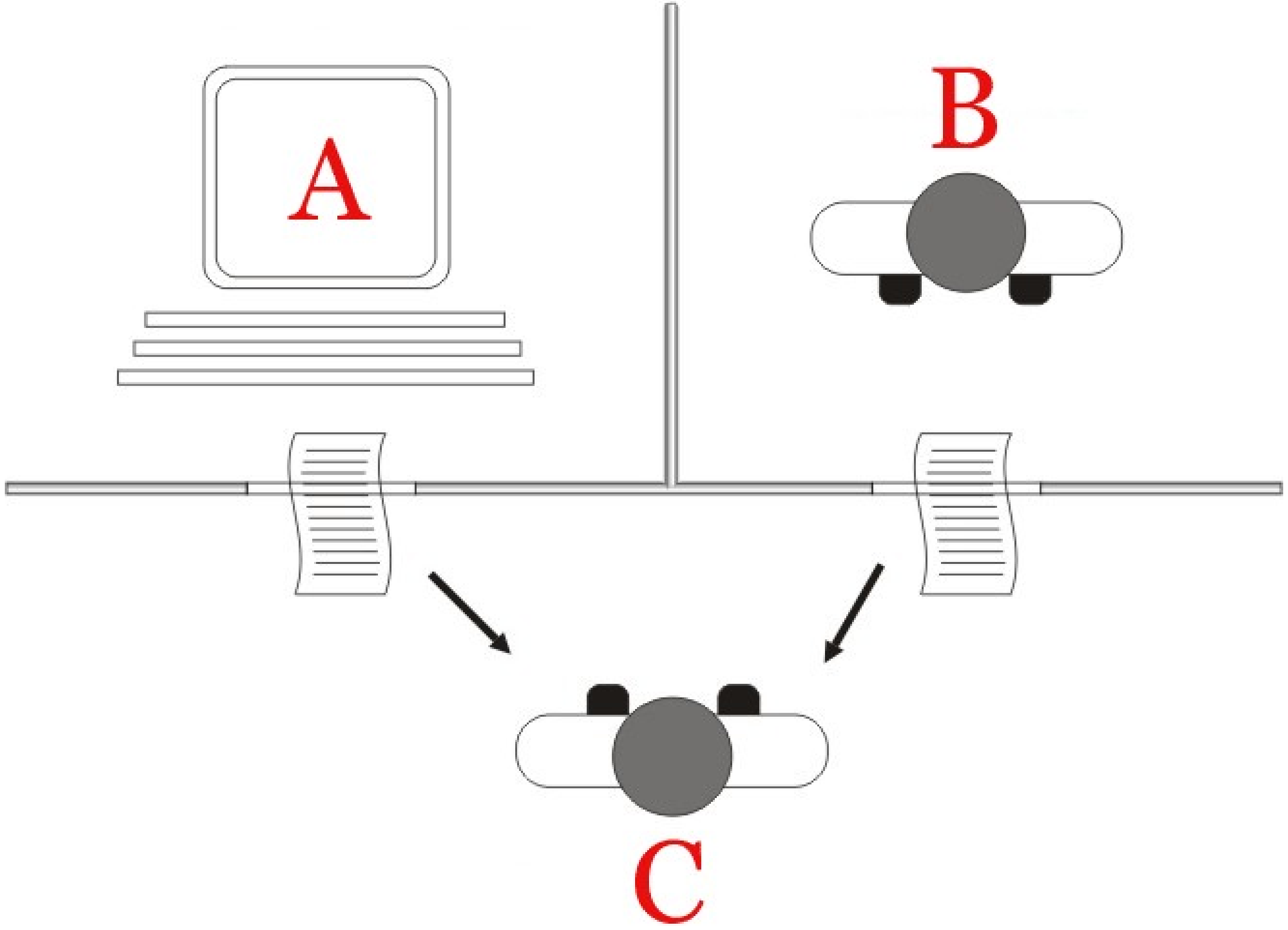
- It is a field of research in computer science that develops and studies methods and software that enable machines to perceive their environment and use learning and intelligence to take actions that maximize their chances of achieving defined goals.

What is AI?

- Approximation based on iterative improvement of operations performed on data based on a mathematical definition of succss

- Maybe by the end, you’ll agree with me.

What is AI? Turing Test

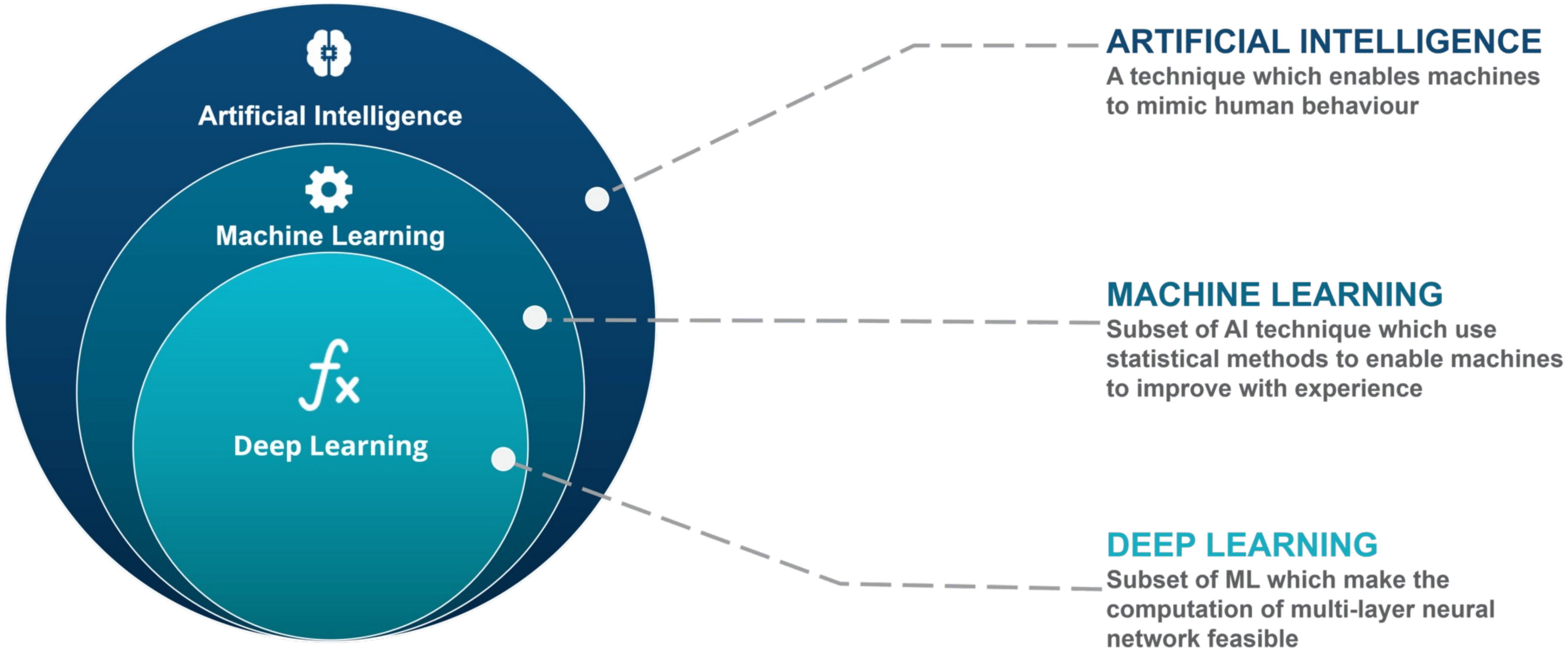


What is AI?

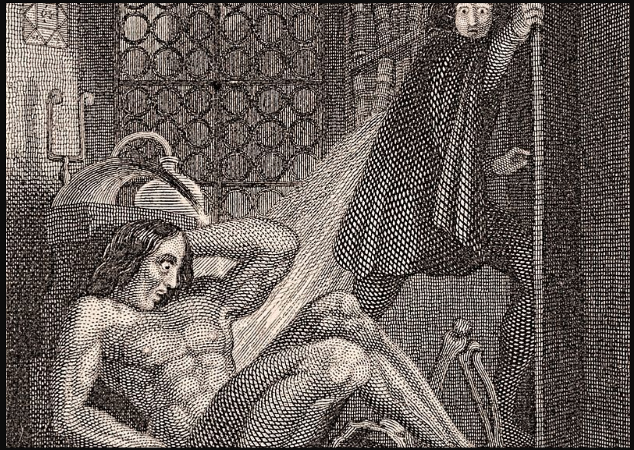
- You know it when you see it?
- No widely accepted definition.

- For the purposes of this class:
  - Any computer assisted decision making system

AI vs ML


Artificial Intelligence
Machine Learning
fx
Deep Learning
ARTIFICIAL INTELLIGENCE
A technique which enables machines to mimic human behaviour
MACHINE LEARNING
Subset of AI technique which use statistical methods to enable machines to improve with experience
DEEP LEARNING
Subset of ML which make the computation of multi-layer neural network feasible



Brief history of AI

- Ancient myths and legends:
    - Talus
    - The Golem
    - Automata
    - Frankenstein

Brief History of AI

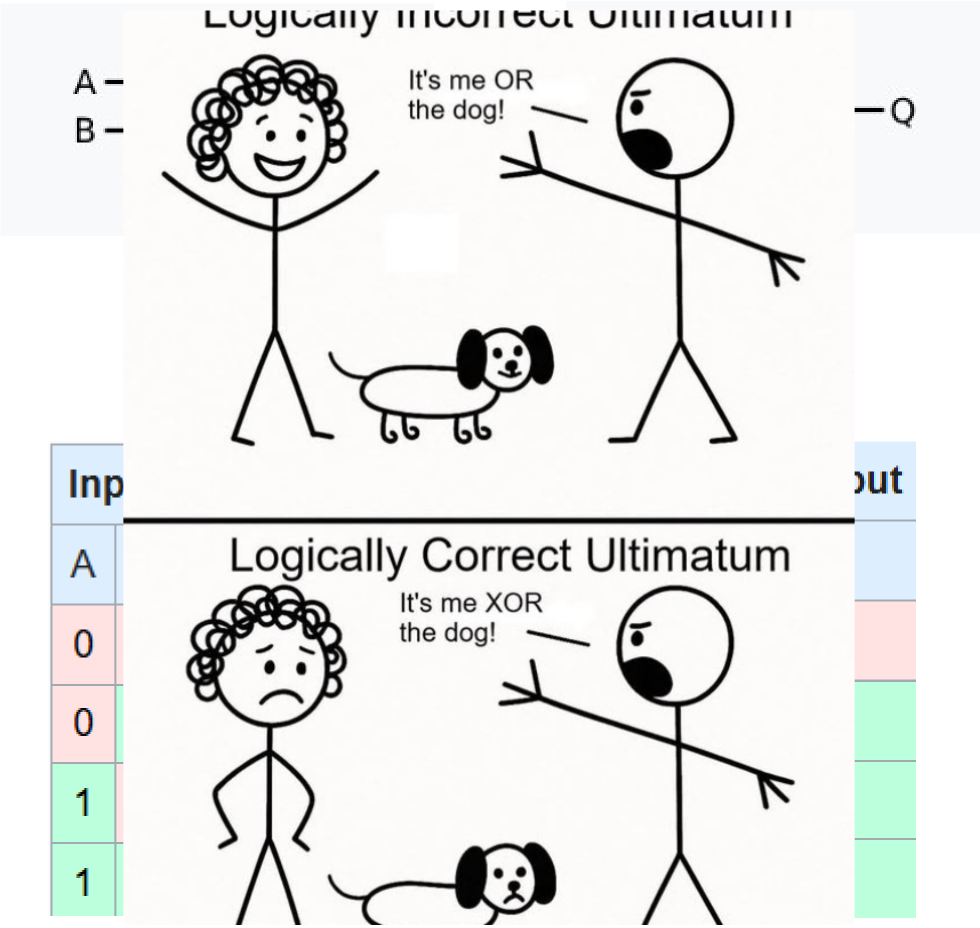
- Great logicians:
  - Al Khwarizmi
  - Leibniz
- Formalization of the laws of logic
  - Boolean logic
- Turing's machine
  - Limitations to machine representation of logic
  - Within that, all problems mathematically representable
  - Outlined the theoretical capabilities of a computer

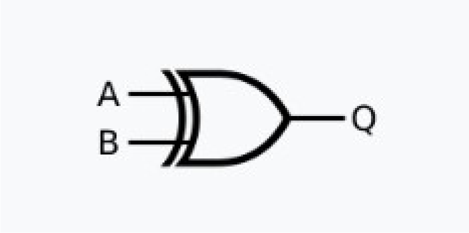
Logical operation	Operator	Notation
Conjunction	AND	$x \wedge y$
Disjunction	OR	$x \vee y$
Negation	NOT	$\neg x$

x	y	$x \wedge y$	$x \vee y$
0	0	0	0
1	0	0	1
0	1	0	1
1	1	1	1

x	$\neg x$
0	1
1	0

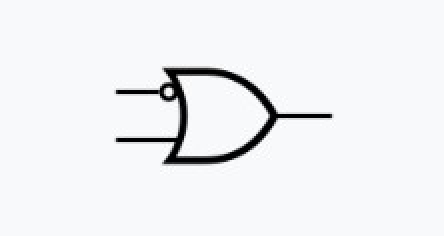
Aside: Logic Gates





XOR

Input		Output
A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

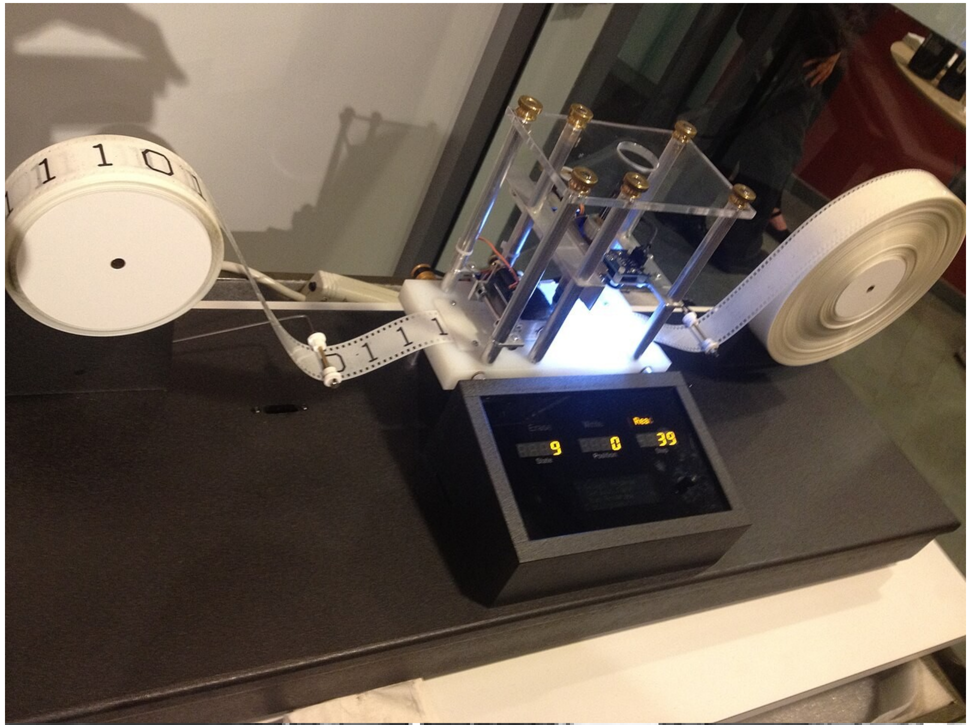


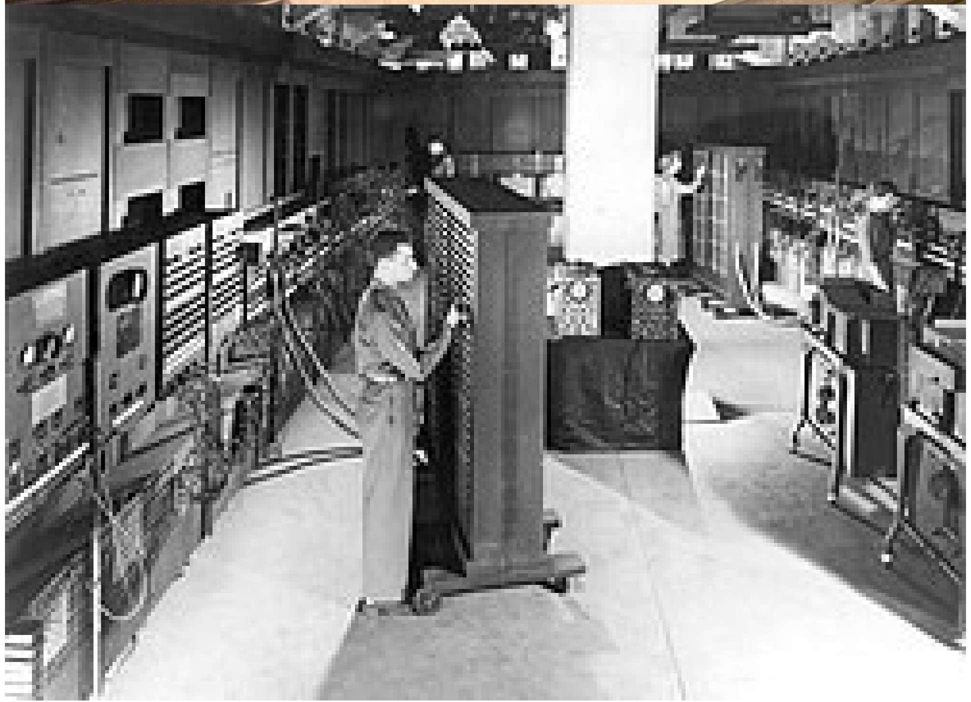
IMPLY

Input		Output
A	B	Q
0	0	1
0	1	1
1	0	0
1	1	1

Brief History of AI

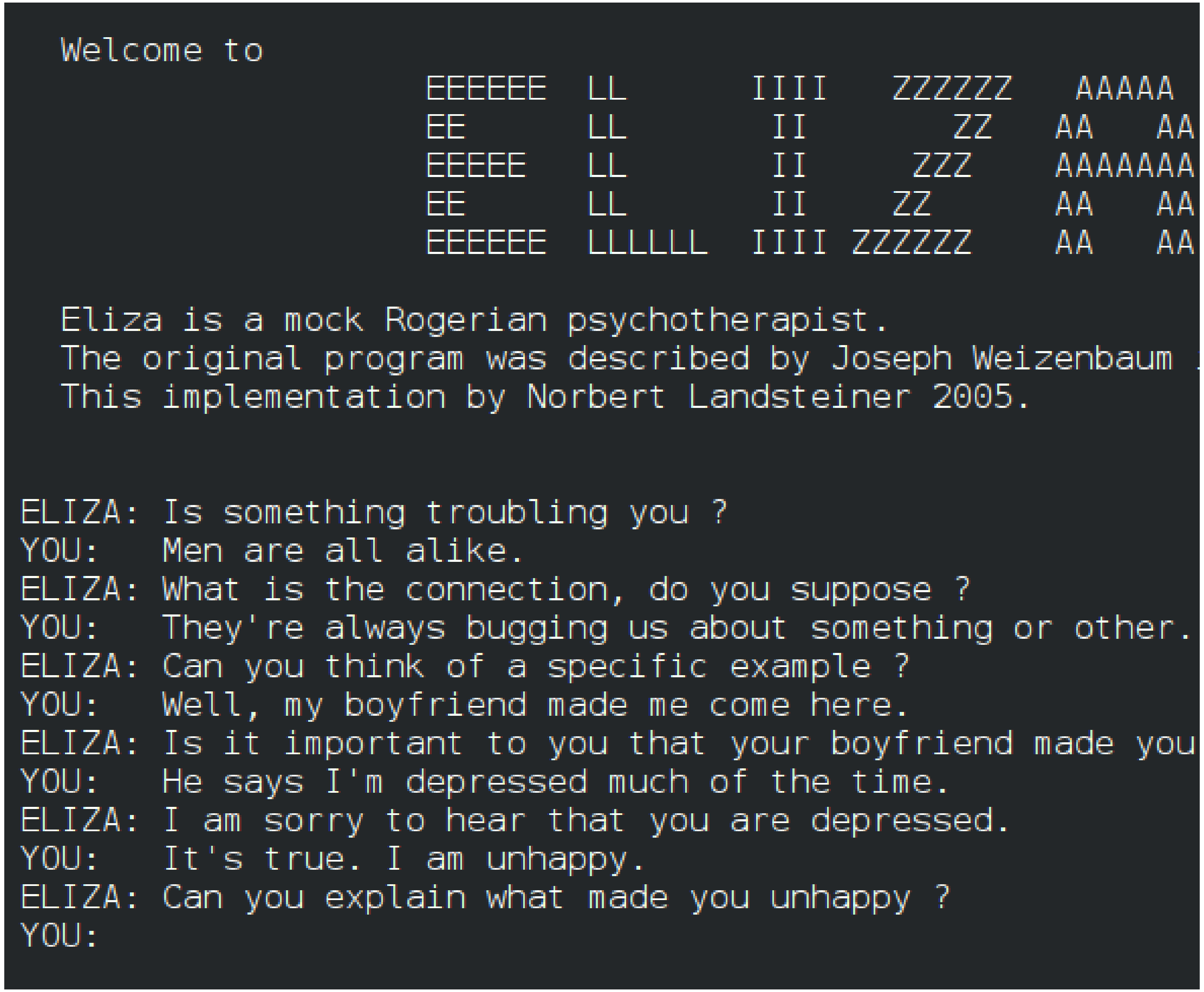
- Realization of Turing's machine
  - Current state, input can perform certain transitions to new state
  - Stored program computer
- Supercomputers in the 1940s
  - Colossal machines meant to replicate Turing's machine
- In the 1960s, first applications to medicine begin





1960s: ELIZA

- First “chatterbot”
- Psychotherapy bot
- Looks for keywords, then flips question back to user
    - Logic to restructure sentences, ask questions based on certain keywords
- First time Turing test was passed



1970’s: MYCIN syster

- 1970’s (Stanford): MYCIN expert
- System for identifying bacteremia
- Proposed a good therapy in ~69% of cases.

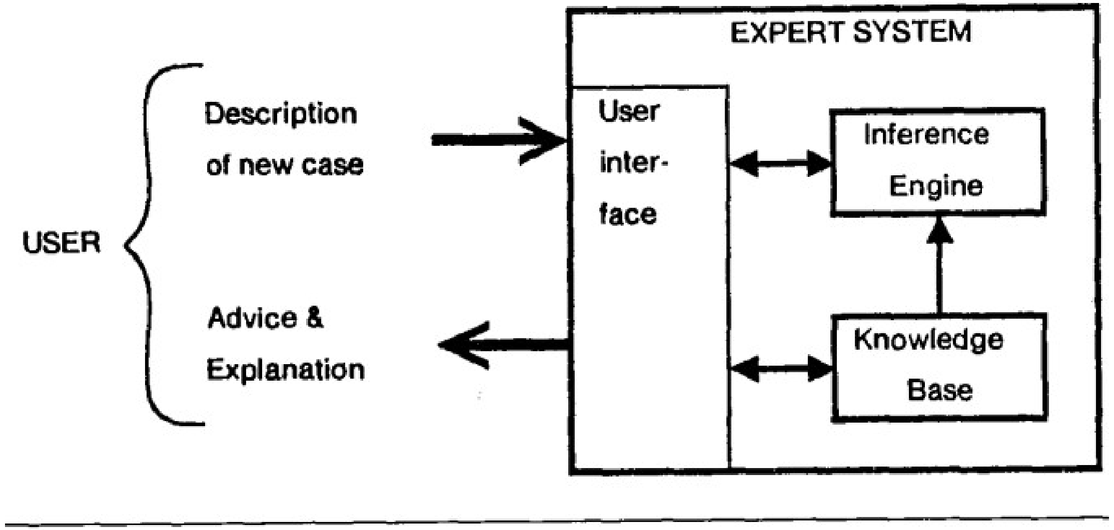


FIGURE 1-1 Major parts of an expert system. Arrows indicate information flow. 4

Dialogue interface

I am ready

** THIS IS A 26 YEAR OLD MALE PATIENT

My understanding is:
The age of the patient is 26
The sex of the patient is male

** FIVE DAYS AGO, HE HAD RESPIRATORY-TRACT SYMPTOMS

What is his name?

** JO

My understanding is:
The name of the patient is Jo
Respiratory-tract is one of the symptoms that the patient had

** A COUPLE OF DAYS BEFORE THE ADMISSION, HE HAD A MALAISE

Please give me the date of admission

** MARCH 12, 1979

My understanding is:
The patient was admitted at the hospital 3 days ago
Malaise is one of the symptoms that the patient had 5 days ago

Thanks to MIT 6.S867 for these slides.

1970’s: MYCIN system

If: Patient with streptococcal bacteremia

If: patient allergic to penicillin

Back to knowledge base

Yes

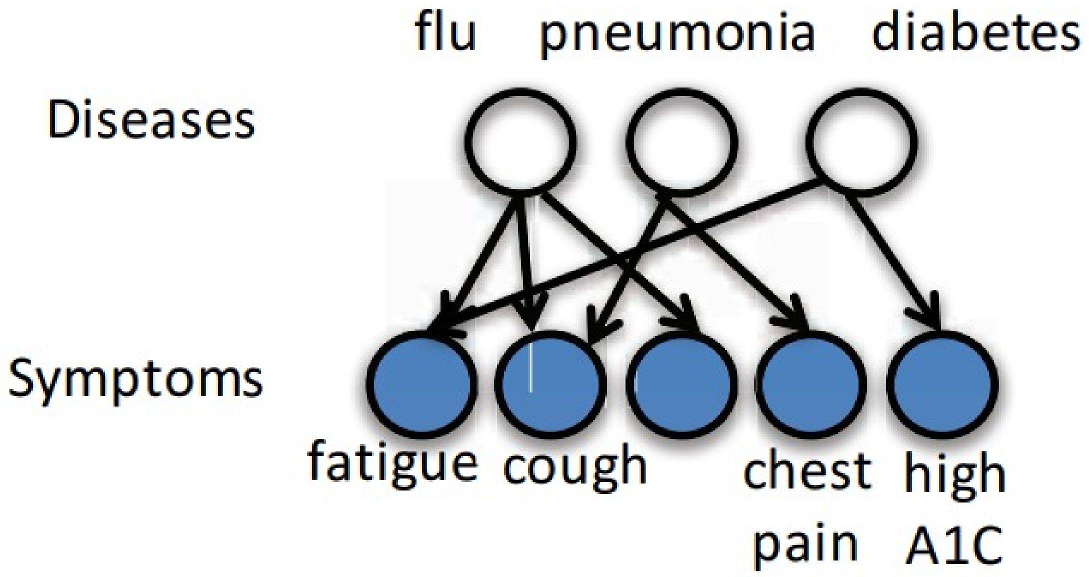
Then: treat with erythromycin or clindamycin

No

Then: treat with Penicillin V

1980s: INTERNIST-1/Quick Medical Reference

- Internal Medicine diagnostic model
- Probabilistic model relating:
    - 570 disease variables
    - 4075 binary (yes/no) symptom variables
    - 45470 connections



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1980’s: The RX Project

PREDNISONE, at a level of 30 mg/day {modal effects}

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usually increases CHOLESTEROL by 50 to 130 mg/dl,
usually increases WEIGHT by 3 to 7 kg,
regularly attenuates NEPHROTIC SYNDROME by 1 to 2 g protein/24 hr,
regularly attenuates GLOMERULONEPHRITIS by 10 to 30% activity,
regularly decreases EOSINOPHILS by 2 to 3% of WBC,
commonly decreases ANTI-DNA by 50 to 90% activity,
occasionally increases GLUCOSE by 20 to 100 mg/dl.

Thanks to MIT 6.S867 for these slides.

Trend

- Notice the trend over time: moving from fixed rules to more probabilistic
- This will continue into 21^{st} century as we move into next lecture

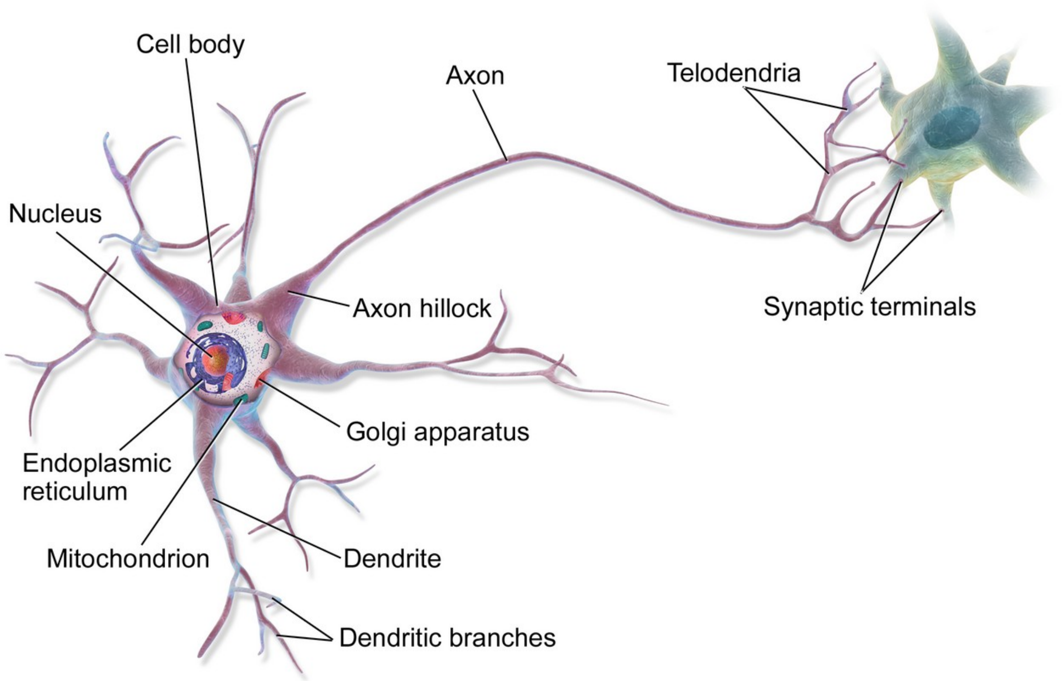
- What does this say about intelligence, knowledge?

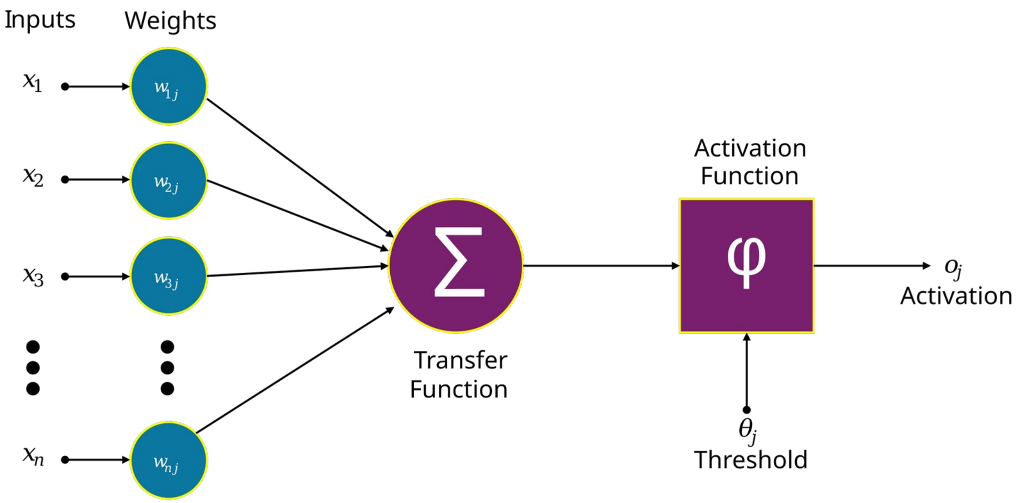
Timeline

Rules	Probabilities	???	???	???
1980s	1990s	2000s	2010s	2020s

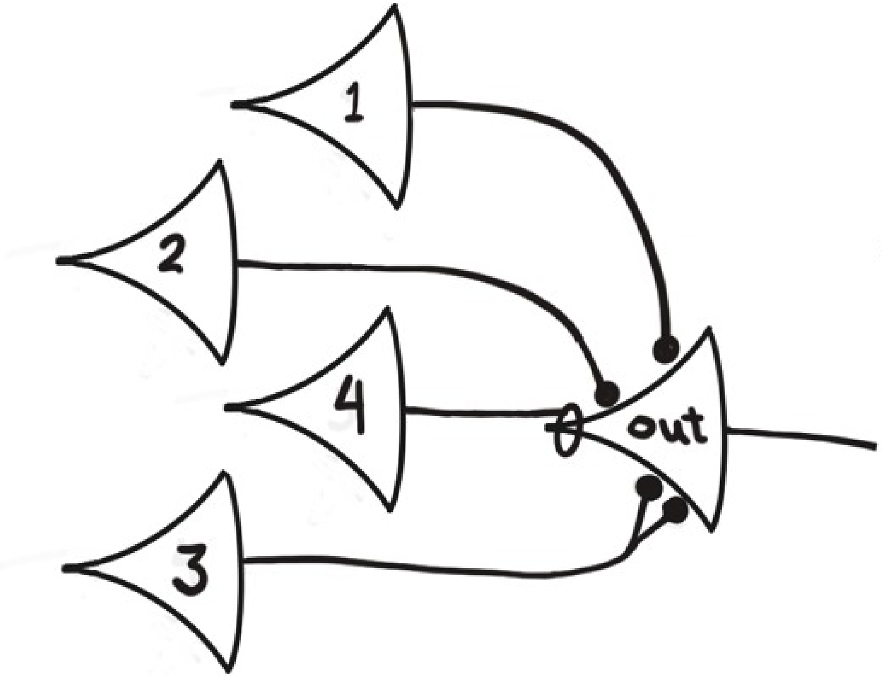
Medicine Inspires AI

- 1940s – The neuron cell is discovered
- Computer scientists wonder – can this be replicated to form the basis of Artificial intelligence?



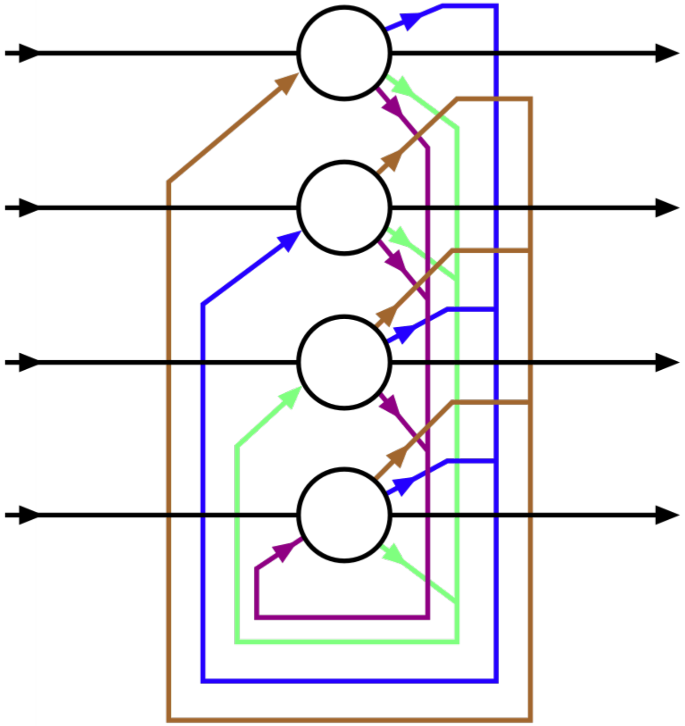


1960s: The Artificial Neural Network



t_i				t_{i+1}
1	2	3	4	out
F	F	F	F	F
T	F	F	F	F
F	T	F	F	F
T	T	F	F	T
F	F	T	F	T
T	F	T	F	T
F	T	T	F	T
T	T	T	F	T
X	X	X	T	F

$$N_{out}(t+1) = ((N_1(t) \cdot N_2(t)) \vee N_3(t)) \cdot \sim N_4(t)$$



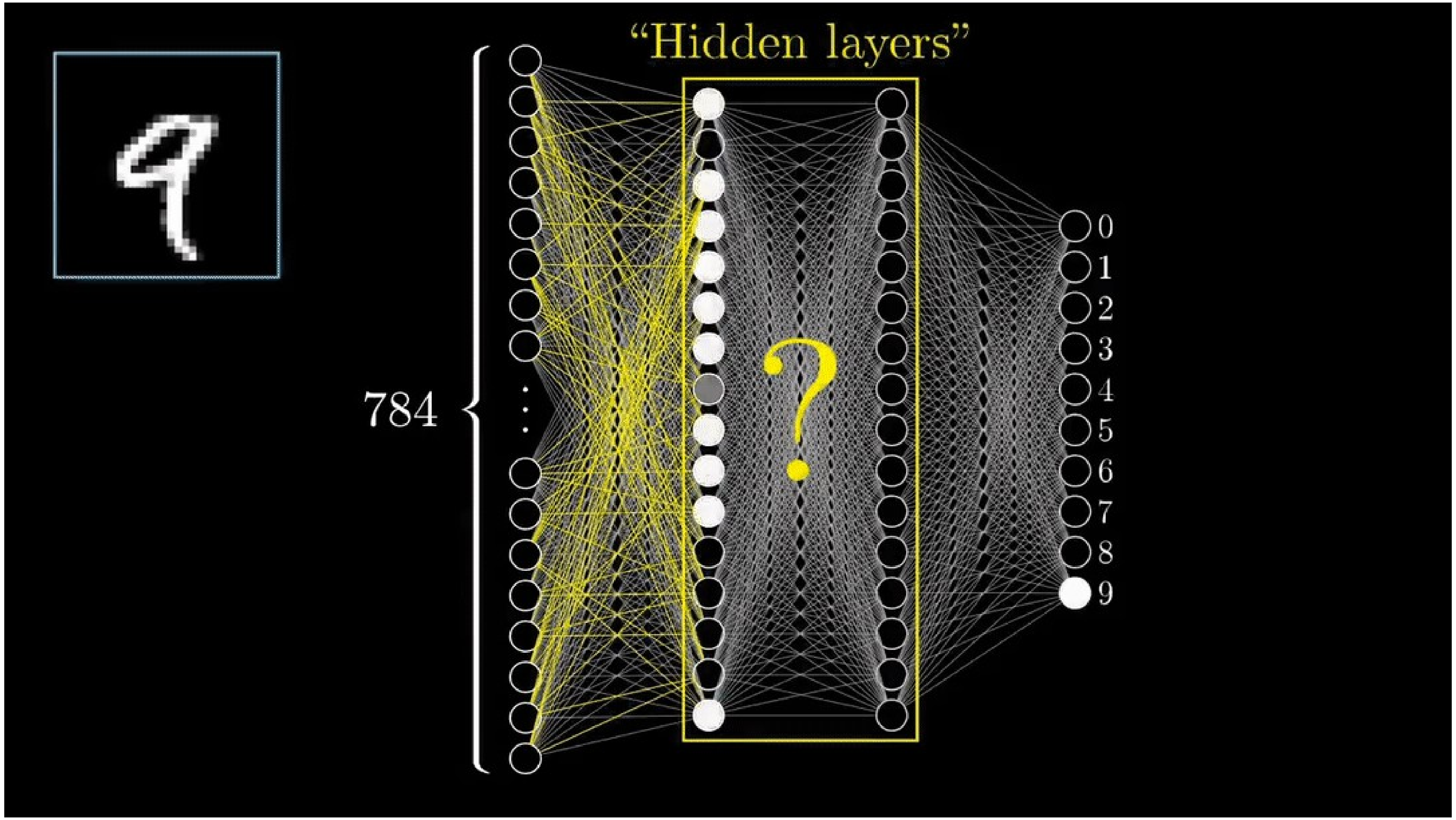
Hopfield network

Limitations

- Cannot model certain functions
- Takes many, many neurons to model
- Sounds great in theory, but not effective in practice
  - How do you set a global learning objective?
  - How do we set a goal and get better at it?

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

1980s/1990s: Backpropagation



We'll pick up here next lecture…

Thanks!